

Breast Cancer among Hispanics, American Indians and Non-Hispanic Whites in New Mexico

MILLICENT EIDSON,^{*} THOMAS M BECKER,[†] CHARLES L WIGGINS,[†] CHARLES R KEY[†] AND JONATHAN M SAMET[†]

Eidson M (Office of Epidemiology, New Mexico Health and Environment Department, Santa Fe, NM, USA), Becker TM, Wiggins CL, Key CR and Samet JM. Breast cancer among Hispanics, American Indians and non-Hispanic whites in New Mexico. *International Journal of Epidemiology* 1994; **23**: 231–237.

Variation in breast cancer occurrence among women in New Mexico's three major ethnic groups has not previously been assessed. To address the descriptive epidemiology of breast cancer in New Mexico Hispanics, American Indians, and non-Hispanic whites, we calculated incidence rates from population-based registry data covering 1969–1987 and mortality data collected from 1958 to 1987. Breast cancer incidence and mortality rates for New Mexico's non-Hispanic white women were comparable to those for white women nationwide. In contrast, American Indian women had extremely low incidence and mortality rates for breast cancer; rates for Hispanics were intermediate, but well below those for non-Hispanic white women throughout the study period. Pronounced temporal trends in breast cancer occurrence were evident among Hispanic women, with the incidence rate increasing by 56% over the 19 years of available data and the mortality rate increasing by nearly 100% over 30 years. Age-specific incidence and mortality rates increased at all ages for successive birth cohorts of Hispanic women. For non-Hispanic whites, increasing incidence and mortality rates were also observed, but the increments were much smaller, approximately 15% for incidence and 30% for mortality. Our data show substantial ethnic differences in breast cancer incidence and mortality in New Mexico, suggesting the need for aetiological investigations to assist in controlling this disease.

Patterns of breast cancer occurrence have been extensively documented in the US. Breast cancer, the most frequent solid tumour in women, has only recently been surpassed by lung cancer as the leading cause of cancer-related deaths among American women.¹ However, the incidence of breast cancer is about three times that of lung cancer and the lifetime probability of developing breast cancer for an American woman is about 10%.² Moreover, data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program indicate that breast cancer incidence rates are rising.³

Breast cancer incidence varies among racial and ethnic groups in the US.⁴ In comparison with non-Hispanic whites, lower incidence rates have been reported for blacks, Asians, American Indians, and Hispanics.^{4,5} The data for American Indian and Hispanic populations

have been extremely limited, however, and temporal trends have not been adequately described for either group. Because New Mexico has the largest percentage of Hispanics and American Indians of all states in the US, we examined the age-adjusted and age-specific breast cancer incidence rates from 1969 to 1987 and mortality rates from 1958 to 1987 in the state to characterize ethnic differences in the trends of breast cancer incidence and mortality.

METHODS

Breast cancer incidence data were obtained from the New Mexico Tumor Registry, a member of the SEER Program. Since 1969, the Registry has ascertained breast and other incident cancer cases statewide.⁶ We obtained coded death certificate data for New Mexico residents from the New Mexico Bureau of Vital Statistics. Cause of death on the certificates was coded according to the seventh revision of the International Classification of Diseases (ICD-7) for the years 1958–1968,⁷ the eighth ICD revision (ICD-8) for the years 1969–1978⁸ and the ninth ICD revision (ICD-9) for the years 1979–1987.⁹ Deaths attributed to female breast cancer were coded as 170–170.9 in ICD-7, and as 174–174.9 in ICD-8 and ICD-9.

^{*} Office of Epidemiology, New Mexico Health and Environment Department, Santa Fe, NM, USA.

[†] Departments of Medicine and Pathology, and the New Mexico Tumor Registry, Cancer Center, University of New Mexico, Albuquerque, NM, USA.

Reprint requests to: Dr Jonathan M Samet, University of New Mexico School of Medicine, New Mexico Tumor Registry, 900 Camino de Salud NE, Albuquerque, NM 87131, USA.

TABLE 1 Average annual age-adjusted^a incidence rates (per 100 000) of breast cancer among New Mexico females, 1969–1987

Time period	Non-Hispanic whites	Hispanics	American Indians
1969–1972	79.6 (938) ^b	36.8 (202)	25.6 (22)
1973–1977	86.9 (1466)	46.8 (363)	19.1 (26)
1978–1982	84.6 (1716)	51.7 (464)	28.7 (50)
1983–1987	93.3 (2234)	57.5 (578)	31.3 (60)

^a Age-adjusted by the direct method to the 1970 US Standard.^b Number of cases.

The New Mexico Tumor Registry determines ethnicity using information from several sources. Hispanics are identified through statements in medical records and on the basis of surname. American Indians are identified by specific designation and by place of residence, and also from records of medical coverage provided by the Indian Health Service. Non-Hispanic whites are people designated as white or Caucasian who do not have a Spanish surname and whose records contain no other indicators of Hispanic ethnicity.

For the mortality data, ethnicity of individuals was assigned by the Bureau of Vital Statistics on the basis of information contained on the death certificates. Hispanic ethnicity was determined on the basis of the decedent's surname, the surnames of the decedent's parents, and specific statements on the death certificate. American Indians were identified solely on the basis of information cited on the death certificate. Non-Hispanic whites were coded as 'white' on the death certificate, and did not have a Spanish surname or other information to indicate that they were Hispanic. We have previously found high validity for the assignment of Hispanic ethnicity by the Bureau of Vital Statistics in comparison with self-reported ethnicity.¹⁰ We also validated the Bureau of Vital Statistics' assignment of American Indian ethnic status by comparing ethnic information collected by the Bureau of Vital Statistics with the New Mexico Tumor Registry data for a sample of 898 American Indians diagnosed with cancer who died from cancer-related or other causes. Tumor Registry records are abstracted primarily from medical records in Indian Health Service facilities. The Indian Health Service clinics and hospitals serve only recognized American Indian patients. We found 95% concordance between the Bureau of Vital Statistics' assignment of American Indian status and the assignment of American Indian race in the Tumor Registry's cancer data base.

Denominators for rate calculations for both incidence of and mortality from breast cancer were derived from

TABLE 2 Average annual age-adjusted^a mortality rates (per 100 000) of breast cancer among New Mexico females, 1969–1987

Time period	Non-Hispanic whites	Hispanics	American Indians
1958–1962	20.3 (245) ^b	10.2 (63)	2.5 (3)
1963–1967	19.6 (287)	11.9 (79)	4.4 (5)
1968–1972	24.0 (355)	13.3 (91)	12.6 (15)
1973–1977	22.1 (377)	14.4 (109)	8.8 (11)
1978–1982	26.8 (559)	19.4 (167)	8.9 (15)
1983–1987	27.5 (640)	18.6 (182)	15.8 (26)

^a Age-adjusted by the direct method to the 1970 US Standard.^b Number of cases.

the censuses of 1960,^{11–13} 1970,^{14–16} and 1980.¹⁷ Since different techniques were used by the US Census Bureau to identify Hispanics in each decennial census, the estimates of the Hispanic population from 1960 to 1980 were adjusted to account for the different enumeration procedures using methodologies previously reported.¹⁰ For the midpoints of the calendar year intervals considered in this analysis, denominators were estimated by interpolation or extrapolation, as appropriate, for those periods not centred on a decennial census. Age-adjusted rates were calculated by the direct method and standardized to the 1970 US standard million population. Case numbers were not sufficient for calculating rates for blacks or other minority populations.

RESULTS

We found differences in incidence and mortality for breast cancer among New Mexico's three major ethnic groups (Tables 1 and 2). A consistent pattern was present for both incidence and mortality rates: American Indians had the lowest rates and non-Hispanic whites the highest, with Hispanics having intermediate rates.

Temporal trends were evident in the rates for Hispanics and non-Hispanic whites. Interpretation of apparent trends in the American Indians is constrained by the small numbers of cases, particularly for the earliest intervals. For non-Hispanic whites, a comparison of the data from 1968–1972 with that of 1983–1987 showed that breast cancer incidence increased by about 15%. Breast cancer mortality also increased in non-Hispanic women (Table 2). The temporal trends were more pronounced among Hispanic women, with consistent increases in incidence and mortality for each successive period of observation. The incidence rate rose by 56% over the 19 years of available data, and the mortality rate nearly doubled over the 30-year period.

Age-specific incidence and mortality rates were calculated for Hispanic and non-Hispanic white women

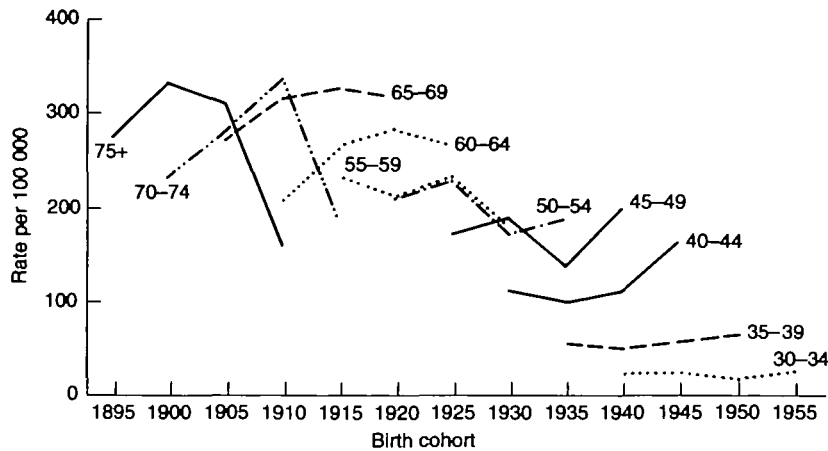


FIGURE 1 Breast cancer age-specific incidence trends by birth cohort for New Mexico non-Hispanic white females, 1969-1987

(Figures 1-4). For the non-Hispanic whites, incidence rates in successive birth cohorts increased for age groups in the premenopausal age range and tended to peak and then decline in the older age groups (Figure 1). In contrast, mortality rates increased for the older age groups in successive birth cohorts and were stable at younger ages (Figure 2). The age-specific incidence rates tended to increase by birth cohort at all ages for Hispanic women (Figure 3) and a similar pattern was evident for mortality rates (Figure 4). For mortality, rates increased sharply for the oldest age groups in successive birth cohorts.

DISCUSSION

We have assessed breast cancer incidence and mortality in Hispanic, American Indian, and non-Hispanic white

women in New Mexico. The Hispanic population in the state includes descendants of the Spanish conquistadors of the 16th century, colonists of the 17th and 18th centuries, and more recent Mexican immigrants;^{18,19} in addition, there was extensive intermarriage of the early settlers with the northern Pueblo Indians. Today, the Spanish language and a common culture and religion characterize this racially mixed group. The American Indian tribes in New Mexico include the Pueblos, the Apaches, and the Navajos; tribal groups that have maintained distinct cultural heritages.

Both genetic and environmental correlates of being Hispanic or American Indian could plausibly influence breast cancer risk. Thus, we found lower incidence and mortality rates of breast cancer in Hispanic and American Indian women compared with non-Hispanic

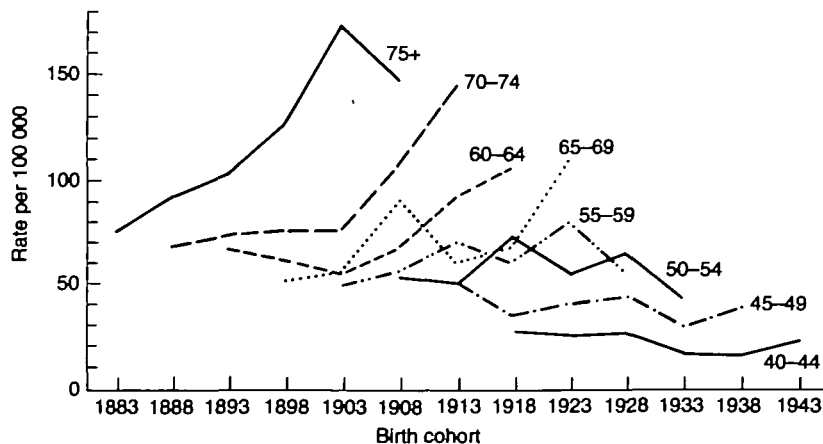


FIGURE 2 Breast cancer age-specific mortality trends by birth cohort for New Mexico non-Hispanic white females, 1958-1987

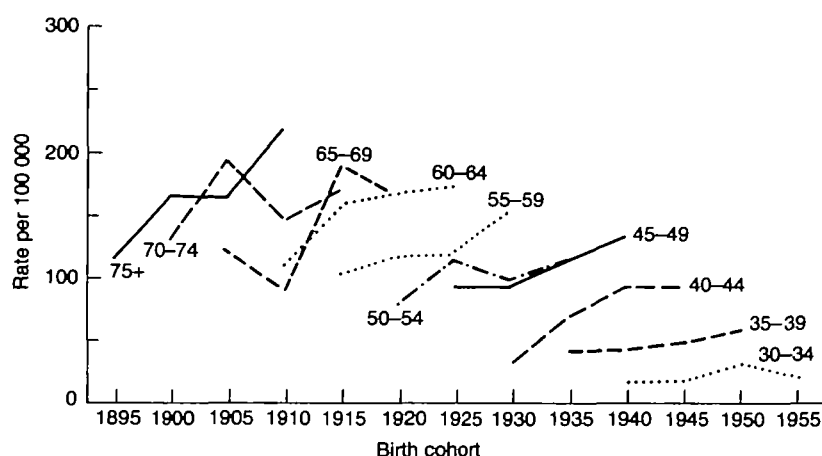


FIGURE 3 Breast cancer age-specific incidence trends by birth cohort for New Mexico Hispanic females, 1969–1987

white women, confirming earlier reports from other populations (Table 3).^{4,5,20–26} The data also document a striking increase in breast cancer occurrence among Hispanic women, beginning as long ago as the 1960s (Tables 1 and 2). The smaller increases of breast cancer incidence and mortality rates among non-Hispanic whites are compatible with data from the SEER Program, which show an increase in breast cancer incidence rates in white women from 89.7 per 100 000 in 1973–1974 to 112.1 in 1986–1987.³ Consistent trends were not evident for American Indian women, but numbers of incident cases and of deaths were small.

Published information on the epidemiology of breast cancer in Hispanic women is extremely limited. Registry

data from Texas,²⁰ California,^{21,22} Denver,⁵ and New Mexico⁶ have indicated a generally lower incidence among Hispanic women (Table 3). Data for New Mexico and Denver provide evidence that breast cancer rates are now increasing in these locations (Tables 1–3). For Denver, data comparing incidence for Spanish-surnamed women for 1969–1971, as documented by the Third National Cancer Survey, with incidence during the period 1979–1981, as documented by the Colorado Cancer Registry, show a 50% increment.⁵ A comparable increase was evident in New Mexico across the same period (Table 1). Increasing rates were not found for Spanish-surname women in Los Angeles County for the years 1972–1982 (Table 3). The increasing rates in New

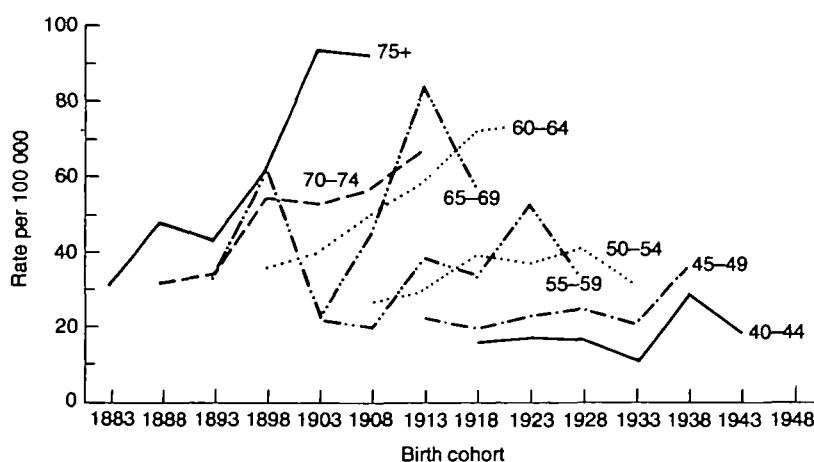


FIGURE 4 Breast cancer age-specific mortality trends by birth cohort for New Mexico Hispanic females, 1958–1987

TABLE 3 Average annual age-adjusted^a incidence rates (per 100 000), white populations

Location	Time period	Non-Hispanic	
		Hispanic	white
El Paso, Texas ^b	1968–1970	57.2	63.9
Los Angeles County, California ^c	1972–1977	65.1	98.8
	1978–1982 ^d	49.0	90.1
Denver, Colorado ^e	1969–1971	39.6	81.6
	1979–1981	69.9	88.8

^aAge-adjusted to the 1970 US population using published age-specific rates.

^bReference 27.

^cReference 28.

^dReference 22.

^eReference 5.

Mexico appear to affect both premenopausal and postmenopausal women (Figures 3 and 4).

Little information, either cross-sectional or longitudinal, is available on the established risk factors for breast cancer in Hispanics. In this predominantly Catholic population, it is likely that first pregnancies occur at younger ages and that parity is high. Some data from Los Angeles and New Mexico support this speculation. For women aged 35–44 and residing in Los Angeles County in 1970, mean parity was 3.5 in Spanish-surnamed and 2.6 in other white women.²⁹ In New Mexico's control series for the Cancer and Steroid Hormones (CASH) study, mean ages at menarche (13.2 years for Hispanics and 13.0 years for non-Hispanic whites) and at first pregnancy (22.7 years for Hispanics and 23.3 years for non-Hispanic whites) were similar, although parity was greater for the Hispanics (4.1 for Hispanics and 2.9 for non-Hispanics) (Dorothy A Pathak, personal communication). In a 1984–1985 survey in an Hispanic community near Albuquerque, we confirmed the overall patterns of the CASH study for high parity of Hispanic women, but age at menarche and at first birth were comparable with data for non-Hispanic white women (Table 4).³⁰ In this population, parity was distinctly lower for women aged 35–44 years.

Studies in the Southwest and nationwide document a higher prevalence of obesity among Hispanics compared with non-Hispanic whites.^{30–33} In a population-based sample of Hispanics in a New Mexico community, the majority of subjects had a body mass index ≥ 25 kg/m², and a substantial proportion exceeded 30 kg/m².³⁰ The prevalence of obesity was much greater in these Hispanics than is shown in nationwide data for US whites. In spite of the greater obesity of Hispanics, two studies in

TABLE 4 Reproductive risk factors (means) in Hispanic females in a community in New Mexico, 1984–1985

	Age at interview (years)		
	18–44	45–64	≥ 65
Age at menarche	12.9	13.0	13.3
Age at first child	20.4	21.5	22.1
Parity	2.5 ^a	5.0	5.8

^aMean = 3.1 for those aged 35 to 44 years.

Texas did not find a greater intake of calories or of fat in a comparison of Hispanics with non-Hispanic whites.^{34,35} In fact, in a study of Mexican Americans on the Texas/Mexico border, the average per cent of calories from total fat among males and females was 34% (R Sue McPherson, personal communication).

Although the data for American Indians suggest possibly increasing rates over the study period, more striking was the extremely low rate of breast cancer in this population. Over the entire period, breast cancer incidence and mortality rates in New Mexico's American Indians were about one-third of the rates for non-Hispanic white women (Tables 1 and 2). Although recent descriptive data on incidence and mortality for breast cancer in other American Indian populations are sparse, the available information points to a similar pattern of low rates. In western Washington from 1975 to 1983, the standardized breast cancer incidence ratio for American Indian women was 0.5 compared with whites.²³ Standardized mortality ratios (SMR) for breast cancer during the 1970s among American Indians in other states were low compared with white women: in North Carolina's American Indians (SMR = 0.4–0.6 depending on time period);²⁵ in the Seneca tribe of New York (SMR = 0.4);²⁴ and in Alaska Natives (SMR = 0.4–0.7 depending on group).²⁶ Mortality data for 1980–1982 collected by the Indian Health Service for American Indians in different parts of the country generally show lower breast cancer mortality rates for American Indians than for all US women.³⁶ However, substantial geographical variation was apparent: SMR varied from 0.1 in the Phoenix Indian Health Service Area to 0.9 in the Nashville Indian Health Service Area.³⁷ Thus, available data suggest lower rates of breast cancer in American Indian women than in the majority population, although the rates are not homogeneous among all American Indian tribes, and age-specific data have not been adequately examined. No published aetiological studies have addressed risk factors for breast cancer in any American Indian population.

Our population-based data have several potential

limitations which we have discussed in detail elsewhere.^{10,37} Potential problems with ethnic misclassification through change of surname at marriage may be greater for Hispanic and non-Hispanic white women than for men.³⁸ In addition, racial misclassification for American Indians (of both sexes) has been shown to be a problem in one SEER registry outside New Mexico.³⁹ Misclassification for New Mexico's American Indians has not been problematic in our own statewide SEER registry,⁴⁰ nor have we found substantial misclassification of American Indian ethnicity in our state mortality data. Data can also be limited by incomplete case ascertainment and by misclassification of cause of death as 'symptoms, signs, and ill-defined conditions'. A substantial number of unattended and unexplained deaths which occur in New Mexico are assigned to the category 'symptoms, signs, and ill-defined conditions'.⁴¹ This category of death is more likely to be assigned to minority women in the state than to the white majority. We do not know whether a larger proportion of unattended deaths caused by breast cancer is assigned to this category for minority women than for non-Hispanic white women.

Despite the potential for bias in our data, however, we have shown ethnic differences in breast cancer incidence and mortality in New Mexico. We have also described the evolving patterns in breast cancer rates, especially among Hispanic women. These descriptive data strongly suggest the need for appropriately designed aetiological investigations, especially among Hispanic women: such information may prove useful in further efforts to control breast cancer in New Mexico.

ACKNOWLEDGEMENT

Partially supported by the Flinn Foundation and by Contract NOI-CN-55426 from the Biometry Branch, National Cancer Institute.

REFERENCES

- Silverberg E, Lubera J. Cancer statistics, 1987. *Cancer* 1987; 6: 41-49.
- American Cancer Society. *Cancer Facts and Figures—1988*. New York: American Cancer Society, 1988.
- National Cancer Institute. In: Gloeckler L A, Hankey B F, Edwards B K (eds). *Cancer Statistics Review 1973-1987*. NIH Publ No. 90-2789. Bethesda, MD: National Cancer Institute, 1990.
- National Cancer Institute. *Cancer among Blacks and Other Minorities: Statistical Profiles*. NIH Publ No. 86-2785. Rockville, MD: National Cancer Institute, 1986.
- Savitz D A. Changes in Spanish surname cancer rates relative to other whites, Denver area, 1969-71 to 1979-81. *Am J Public Health* 1986; 76: 1210-15.
- Key C R. Cancer incidence and mortality in New Mexico, 1973-1977. I. In: Young, Jr. J L, Percy C L, Asire A J (eds). *Surveillance, Epidemiology and End Results. Incidence and Mortality Data, 1973-1977*. National Cancer Institute, Monograph No. 57. Washington, DC: Government Printing Office, 1981, pp. 489-595.
- World Health Organization. *Manual of the International Statistical Classification of Disease, Injuries, and Causes of Death*. Based on the Recommendations of the Seventh Revision Conference, 1955. Geneva: WHO, 1957.
- World Health Organization. *Manual of the International Statistical Classification of Disease, Injuries, and Causes of Death*. Based on the Recommendations of the Eighth Revision Conference, 1965. Geneva: WHO, 1967.
- World Health Organization. *Manual of the International Statistical Classification of Disease, Injuries, and Causes of Death*. Based on the Recommendations of the Ninth Revision Conference, 1975. Geneva: WHO, 1977.
- Samet J M, Wiggins C L, Key C R, Becker T M. Mortality from lung cancer and chronic obstructive pulmonary disease in New Mexico, 1958-82. *Am J Public Health* 1988; 78: 1182-86.
- United States Bureau of the Census. *Census of the Population, 1960. General Population Characteristics. Final Report. New Mexico. PC(1)-33B*. Washington, DC: Government Printing Office, 1961.
- United States Bureau of the Census. *Census of the Population, 1960. Subject Reports: Persons of Spanish Surnames. Final Report. New Mexico. PC(2)-1B*. Washington, DC: Government Printing Office, 1963.
- United States Bureau of the Census. *Census of the Population, 1960. Subject Reports: Non-White Persons by Race. Final Report. New Mexico. PC(2)-1C*. Washington, DC: Government Printing Office, 1963.
- United States Bureau of the Census. *Census of the Population, 1970. General Population Characteristics. Final Report. New Mexico. PC(1)-B33*. Washington, DC: Government Printing Office, 1971.
- United States Bureau of the Census. *Census of the Population, 1970. General Social and Economic Characteristics. Final Report. New Mexico. PC(1)-C33*. Washington, DC: Government Printing Office, 1971.
- United States Bureau of the Census. *Census of the Population, 1970. Subject Reports. Final Report. New Mexico. PC(2)-C1F*. Washington, DC: Government Printing Office, 1973.
- United States Bureau of the Census. *Census of the Population, 1980. General Population Characteristics. Final Report. New Mexico. PC80-1-B33*. Washington, DC: Government Printing Office, 1981.
- Gonzales N L. *The Spanish-Americans of New Mexico: A Heritage of Pride*. Albuquerque, NM: University of New Mexico Press, 1969.
- Fergusson E. *New Mexico: A Pageant of Three Peoples*. Albuquerque, NM: University of New Mexico Press, 1973.
- McDonald E J, Heinze E D. *Epidemiology of Cancer in Texas: Incidence Analyzed by the Type, Ethnic Group, and Geographic Location*. New York: Raven Press, 1978.
- Menck H R, Henderson B E. Cancer incidence rates in the Pacific Basin. *National Cancer Institute Monograph*. 1979; 53: 119-24.
- Muir C, Waterhouse J, Mack T et al. (eds). *Cancer Incidence in Five Continents, Vol. 5*. Lyon, International Agency for Research on Cancer, 1987, pp. 307-09.
- Norsted T L, White E. Cancer incidence among Native Americans of Western Washington. *Int J Epidemiol* 1989; 18: 22-27.
- Mahoney M C, Michalek A M, Cummings K M et al. Cancer mortality in a northeastern Native American population. *Cancer* 1989; 64: 187-90.
- Horner R D. Cancer mortality in Native Americans in North Carolina. *Am J Public Health* 1990; 80: 940-44.
- Lanier A P, Bulkow L R, Ireland B. Cancer in Alaskan Indians, Eskimos, and Aleuts, 1969-1983. Implications for etiology and control. *Public Health Leaflets* 1989; 104: 658-64.

- ²⁷ Waterhouse J, Muir C, Correa P. *Cancer Incidence in Five Continents, Vol. 3*. Lyon: International Agency for Research on Cancer, 1976 pp. 233–39.
- ²⁸ Waterhouse J, Muir C, Shanmugaratnam K *et al.*: *Cancer Incidence in Five Continents, Vol. 4*. Lyon: International Agency for Research on Cancer, 1982, pp. 301–07.
- ²⁹ Mack T, Menck H. Epidemiology of cancer of the gallbladder and extra-hepatic biliary passages I. In: Correa P, Haenszel W. (eds) *Epidemiology of Cancer of the Digestive Tract*. The Hague: Nijhoff, 1982, pp. 227–42.
- ³⁰ Samet J M, Coultas D B, Howard C A, Skipper B J, Hanis C L. Diabetes, gallbladder disease, obesity, and hypertension among Hispanics in New Mexico. *Am J Epidemiol* 1988; **128**: 1302–11.
- ³¹ Stern M P, Gaskill S P, Allen C R Jr *et al.* Cardiovascular risk factors in Mexican Americans in Laredo, Texas. I. Prevalence of overweight and diabetes and distributions of serum lipids. *Am J Epidemiol* 1981; **113**: 546–55.
- ³² Mueller W H, Joos S K, Hanis C L *et al.* The diabetes alert study: growth, fatness, and fat patterning, adolescence through adulthood in Mexican Americans. *Am J Phys Anthropol* 1984; **64**: 389–99.
- ³³ National Center for Health Statistics. Prevalence of overweight for Hispanics — United States, 1982–1984. *MMWR* 1989; **38**: 838–42.
- ³⁴ Haffner S M, Knapp J A, Hazuda H P *et al.* Dietary intakes of macronutrients among Mexican Americans and Anglo Americans: The San Antonio heart study. *Am J Clin Nutr* 1985; **42**: 1266–75.
- ³⁵ Newell G R, Borud L G, McPherson R S *et al.* Nutrient intakes of whites, blacks, and Mexican Americans in Southeast Texas. *Prev Med* 1988; **17**: 622–33.
- ³⁶ United States Congress, Office of Technology Assessment. *Indian Health Care, OTA-H-290*. Washington, DC: Government Printing Office, April 1986.
- ³⁷ Becker T M, Wiggins C L, Key C R, Samet J M. Ischemic heart disease mortality in Hispanics, American Indians, and non-Hispanic whites in New Mexico, 1958–1982. *Circulation* 1988; **78**: 302–09.
- ³⁸ Wiggins C L, Becker T M. Methods. In: Becker T M, Wiggins C L, Elliott R S, Key C R, Samet J M (eds). *Racial and Ethnic Patterns of Mortality in New Mexico*. Albuquerque, NM: University of New Mexico Press, 1993.
- ³⁹ Frost F, Taylor V, Fries E. Racial misclassification of Native Americans in a surveillance, epidemiology, and end results cancer registry. *J Natl Cancer Inst* 1992; **84**: 957–62.
- ⁴⁰ Valway S. Racial Misclassification of American Indians in a Southwestern SEER registry: Comparisons with Indian Health Service Data. (Abstract.) *Cancer Update*, MD Anderson Cancer Center, Houston, Texas, February, 1991.
- ⁴¹ Becker T M, Wiggins C L, Key C R, Samet J M. Symptoms, signs, and ill-defined conditions: A leading cause of death among minorities. *Am J Epidemiol* 1990; **131**: 664–68.

(Revised version received August 1993)